Augmented GRU with one-head retrieval

Three GRU layers of 32 units | 128-unit prediction head | 46,257 parameters | representative seed 3

A Encode the inputs available before response t

3 timing values + availability flag
Recency; item age; prior mean spacing
4 numeric coordinates

4 LKT predictors
Intercept; student logitdec;
student-item logitdec; student-item PPES

Current item ID
144 known items

Current skill ID
36 known skills

Time encoder
Linear 4 → 32; ReLU
Linear 32 → 16
16 learned time values

History-feature encoder
Linear 4 → 32; ReLU
Linear 32 → 16
16 learned history values

Learned item embedding
145 × 16 table
144 known + 1 reserved zero row
16 item values

Learned skill embedding
37 × 16 table
36 known + 1 reserved zero row
16 skill values

Concatenate 16 time + 16 history + 16 item + 16 skill = 64 values
Input dropout 0.1 produces x(t), shared by retrieval, prediction and memory update

Recency = time since the previous response (any item). Item age and prior spacing use this student-item exposure history, including studies.

B Retrieve earlier memories: one head, width 32

Current event e(t): 96 values
Previous h3(t−1): 32 + x(t): 64

Past completed events from this participant
Store e(j): 96 and post-answer h3(j): 32; newest 100 events, j < t

Shared Q/K linear 96 → 32
Query q(t): 32 values
Uses current e(t)

Shared Q/K linear 96 → 32
Key k(j): 32 values per event
Uses past e(j)

Value linear 32 → 32
v(j): 32 values per event
Uses past post-answer h3(j)

Scores q(t) · k(j) / sqrt(32)
Mask to earlier 100 events; softmax
One weight a(t,j) per past event

Weighted sum: Σ a(t,j) v(j), 32 values
Output linear 32 → 32
Retrieved memory r(t): 32 values
Empty history gives r(t) = 0

Q and K share weights and bias. All projections have bias. No attention decay or dropout.

C Predict the current response

Retrieved r(t): 32 + current x(t): 64
Concatenate 96 values

Linear 96 → 128
ReLU
Head dropout 0.1

Linear 128 → 1
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10^{-5}

The current answer is still unknown. Retrieved memory replaces the direct h3(t−1) input to this prediction head.

D Observe the answer, update all three GRU layers, then write the bank

Observed response y(t)
0 = incorrect; 1 = correct

Response encoder
Linear 1 → 32; ReLU; linear 32 → 16
16 learned answer values

Current x(t): 64 + encoded y(t): 16
80 values into GRU layer 1

Previous h1(t−1)
32 state values

Previous h2(t−1)
32 state values

Previous h3(t−1)
32 state values

GRU layer 1
32 units
h1(t): 32 values

Dropout
0.1

GRU layer 2
32 units
h2(t): 32 values

Dropout
0.1

GRU layer 3
32 units
h3(t): 32 values

Keep all three GRU states for the next trial
h1(t), h2(t), h3(t): 3 × 32 values

For the next trial: append [e(t) from B: 96, new h3(t): 32]
Keep the newest 100 completed event records; e(t) is pre-answer, h3(t) is post-answer

All encoders, embeddings, GRU layers, retrieval projections and prediction weights learn jointly from response-prediction loss.

At a new participant: zero GRU states and an empty bank. The bank read feeds prediction; the GRU update consumes x(t) and encoded y(t).

Timing values use log(1 + seconds) and training-only scaling. Undefined prior spacing stays missing in the data, with an availability flag.

Only completed scored responses enter the bank. Dropout is off during evaluation. States and bank carry forward; gradients detach at 100-event BPTT chunk boundaries.